

**BEFORE THE PUBLIC UTILITIES COMMISSION  
OF THE STATE OF CALIFORNIA**



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Order Instituting Rulemaking to  
Continue the Development of Rates  
and Infrastructure for Vehicle  
Electrification.

Rulemaking 18-12-006  
(Filed December 13, 2018)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON  
THE ASSIGNED COMMISSIONER'S RULING REGARDING  
IMPLEMENTATION OF ASSEMBLY BILL 841**

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February 5, 2021

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission” or “CPUC”), the Vehicle-Grid Integration Council (“VGIC”) <sup>1</sup> hereby submits these opening comments on the Assigned Commissioner’s Ruling Regarding Implementation of Assembly Bill (“AB”) 841, issued on January 15, 2021.

**I. Introduction**

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of electric vehicles (“EV”) and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from EV deployments and flexible EV charging and

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<sup>1</sup> VGIC member companies include American Honda Motor Co., Inc., Connect California LLC, Enel X North America, Inc., Fermata, LLC., Fiat Chrysler Automobiles, Ford Motor Company, General Motors Company, Nissan North America, Inc., Nuvve Corporation, The Mobility House, Toyota Motor North America, Inc., and Ossiaco, Inc. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all of the individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>).

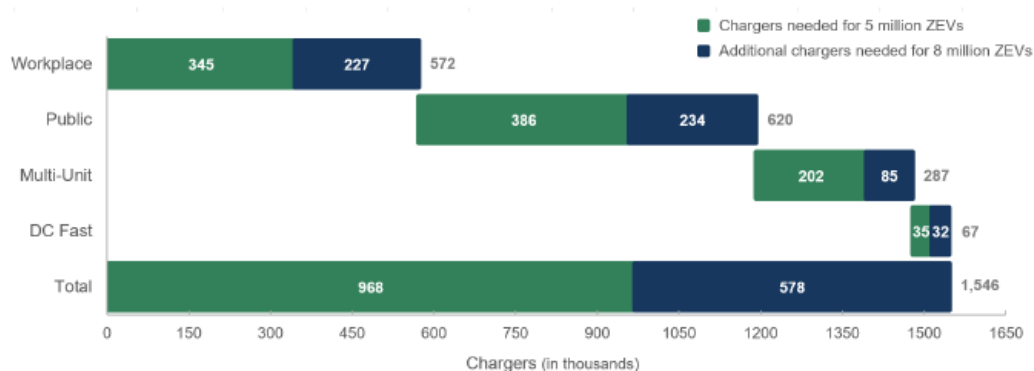
discharging is recognized and compensated in support of achieving a more reliable, affordable, and efficient electric grid.

## **II. Relationship of AB 841 to Overarching TE Issues and ALM**

In considering how to implement AB 841, the Commission should bear in mind two critical facts regarding California's overall efforts to electrify the transportation sector.

- *Fact 1: There is a significant gap between the chargers currently projected to be deployed and the number of chargers needed to reach California's 2025 and 2030 EV goals.* The California Energy Commission ("CEC") recently issued its Electric Vehicle Charging Infrastructure Assessment in response to AB 2127. The report showed that while there are over 188,000 chargers installed or projected to be installed by 2025, this still leaves a gap of about 62,000 chargers needed to meet the state's goals for 1.5 million EVs by 2025. Moreover, there is a gap of over 1.5 million chargers to meet state goals for 8 million EVs by 2030. Absent significant additional publicly funded chargers, many of these will likely need to come from private installations.

**Figure 1: Projected 2030 Charger Counts to Support 5 Million and 8 Million Light-Duty Zero-Emission Vehicles**



Models project that California will need 968,000 shared private and public chargers in 2030 to support 5 million ZEVs, and over 1.5 million chargers to support 8 million ZEVs. Counts for chargers at workplaces, public destinations, and multi-unit dwellings generally indicate the number of Level 2 chargers needed. In some cases, Level 1 chargers may be sufficient at select public destinations primarily serving transportation network company vehicles and at select multi-unit dwellings. These values do not include chargers at single family residences.

Source: CEC and National Renewable Energy Laboratory

- Fact 2: Ensuring a positive experience for new EV drivers and site hosts is central to accelerating EV adoption in support of electrification goals.* Meeting California’s ambitious transportation electrification goals will be made much more difficult without robust demand for EV adoption relative to internal combustion engines (“ICE”). This in turn means that the Commission should ensure that EV-related programs and policies are able to accommodate a wide range of customer needs and preferences and are designed with the customer experience in mind. This concept applies to both new EV drivers who may be hesitant about the longer refueling times required for EVs, as well as new EV supply equipment (“EVSE”) site hosts who could be deterred from installing charging stations if they encounter unexpected cost barriers.

With respect to the gap in chargers (as well as the customer experience), AB 841 helps to overcome this barrier by removing site host cost responsibility for upfront costs related to new or upgraded utility service connections that, if not covered by transportation electrification (“TE”) program funding, may otherwise be prohibitive. As was discussed by Peninsula Clean Energy in

the January 29<sup>th</sup> Automated Load Management (“ALM”) workshop, some site hosts have been deterred from installing EVSE due to unforeseen distribution upgrade costs that made the projects too expensive to move forward. However, VGIC also recognizes that this presents a new challenge for encouraging ALM and ensuring associated benefits to ratepayers. Under AB 841, there is no direct incentive for site hosts to pursue load management solutions if they are not responsible for utility-side distribution upgrade costs. Meanwhile, allowing virtually unrestricted distribution upgrade costs for EVSE deployment could place a potentially significant cost burden on all other ratepayers with little means to mitigate those costs.

One hypothetical solution to this challenge (which VGIC does not support, but is implicated in the Assigned Commissioner Ruling (“ACR”)) would be to impose new load management requirements (e.g. via ALM) on any ratepayer funded TE infrastructure, including those covered by AB 841. However, VGIC is concerned that this approach would risk both: 1) failing to close the gap in charger deployment since prospective site hosts could be subject to, and deterred by, additional upgrade costs if they do not wish to install ALM, and 2) failing to accommodate customers with faster or higher capacity charging, needs thus jeopardizing overall EV adoption rates. Moreover, we do not view this approach as consistent with the language of AB 841 overall. Instead, ALM should be an option in all EVSE installations, but not a requirement. Similarly, there should not be use cases that are automatically considered appropriate or inappropriate for ALM. Those decisions should be left to the customer if they decide their charging needs can be met using ALM. A good example is DC fast charging at destination charging centers. Most likely, EV customer needs at such a location will require an immediate, very fast, full charge that will in turn require that the full nameplate of the EVSEs be

used and ALM is an unfavorable choice. However, if a charging provider believes they can provide this service to the satisfaction of their intended customers using a sub-nameplate connection with ALM, they should be free to choose to do so. We reiterate however, that ALM should not be a requirement under AB 841, and providers should not be forced to accept a smaller than nameplate connection (whether explicitly required or implicitly required due to cost barriers) if they do not wish to do so. As an alternative, VGIC proposes that the Commission could require each investor-owned utility (“IOU”) to develop a new “shared savings” incentive mechanism as an option for customers that is designed to encourage (but not require) site hosts to incorporate ALM. If implemented successfully, this could provide sufficient flexibility to EV customers and site hosts that it would avoid slowing down transportation electrification, while also reducing overall costs borne by ratepayers from EV-related distribution upgrades under AB 841 by encouraging load management. A conceptual version this approach was presented by VGIC at the January 29<sup>th</sup>, 2021 ALM workshop, which was informed by discussion with the Joint Commenters<sup>2</sup> as represented by the informal comments provided by the Joint Commenters on January 15<sup>th</sup>, 2021.

### **III. VGIC Proposed Optional ALM Incentive Mechanism**

Under VGIC’s proposed approach we envision that the utility-side distribution upgrades covered by ratepayers under AB 841 would normally be based on the maximum charging level required to meet the desired charging needs of a site host installing new EVSE equipment. For example, if a multi-unit dwelling (“MUD”) seeks to deploy a charging station with 10 ports, the

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<sup>2</sup> Environmental Defense Fund, Enel X, Nuve, Siemens, Greenlots, the Vehicle-Grid Integration Council, Advanced Energy Economy, and the Natural Resources Defense Council (collectively, “Joint Commenters”)

distribution upgrades would be sized to accommodate a new service with 10 ports charging at full load. Assuming 8 kW per port, that would equate to 80 kW of incremental coincident charging demand. This is illustrated in the Figure below with the first blue bar on the left meant to depict this as the “baseline” level of distribution upgrade, and the associated costs borne by ratepayers (note that these are round numbers used for illustrative purposes and may not reflect a real-world situation).

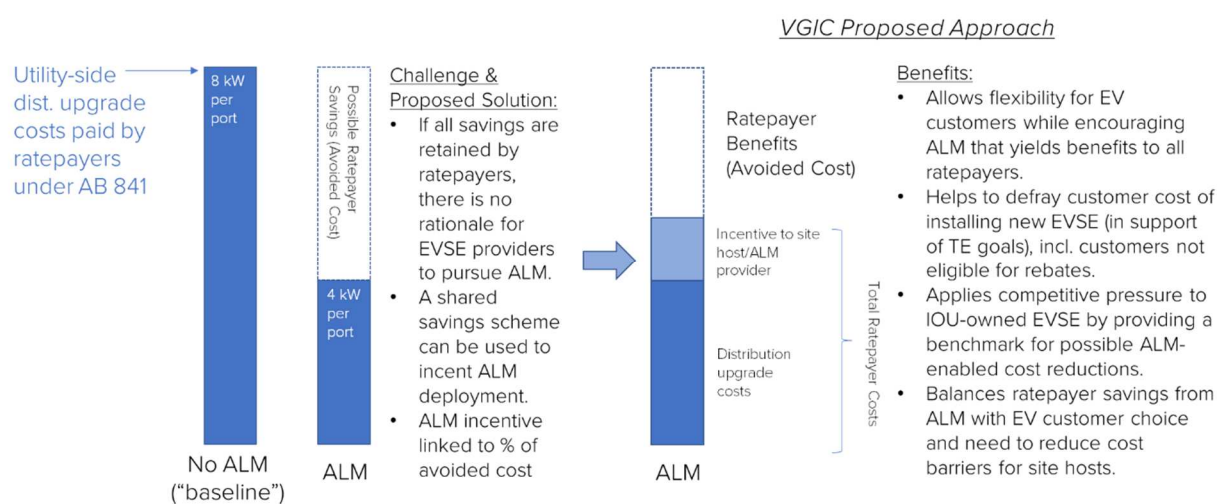


Figure 1. Overview of VGIC's proposed approach (numbers used for illustrative purposes and may not reflect a real-world situation)

The second bar shows what might be practically achievable through ALM. For example, perhaps the distribution upgrades could be reduced from what is required for 10 ports to the equivalent of 5 ports through ALM thus reducing coincident charging demand to 40 kW, or 4 kW per port (on average). Using ALM, customers would not be restricted to only charging at maximum of 4 kW per port. Rather, some EVs may be charging at 8 kW, while others who do not immediately need a charge can delay charging entirely, or charge at a lower rate. The blue bar is reduced accordingly, and the hollow portion on top would be the savings realized by utility ratepayers. Under a strict interpretation of AB 841, there is no rationale or incentive for the site

host or EVSE provider to incorporate ALM since there is no way for the site host to directly capture the benefits of these avoided upgrade costs. Meanwhile, the third bar on the right shows how a shared savings mechanism could work to help encourage ALM deployment on a voluntary basis, with an incentive provided to the site host or ALM provider (shown in light blue). The level of the incentive would be linked to a percentage of avoided distribution upgrade costs realized by implementing ALM. VGIC believes there are many benefits to this approach including the following:

- It allows flexibility for EV customers to choose what level of charging service they need without fear of unexpected costs, while still encouraging ALM that yields benefits to all ratepayers.
- It supports achievement of TE goals by helping to defray customer cost of installing new EVSE, including customers not otherwise eligible for other rebates.
- It applies competitive pressure to IOU-owned EVSE and make-ready investments by providing a benchmark for achievable ALM-enabled cost reductions.
- It balances ratepayer savings from ALM with EV customer choice and need to reduce cost barriers for site hosts.

VGIC proposes that the incentive could be comprised of both an upfront payment component for non-demand related costs that are readily known in advance without extensive engineering studies (e.g. meter, service drop). These would primarily apply to customers using ALM to enable EV use on an existing service. In addition, there could be an on-going payment (e.g. through a bill credit) reflecting an estimated average of avoided demand-related costs (e.g. transformer upgrades) achieved by all ALM customers on a specific circuit. This could apply to



customers using ALM on both existing or new service and could scale with the amount of ALM applied (versus nameplate charging potential). The bill credit mechanism has the added advantage of providing greater certainty to new EVSE customers by not requiring them to await lengthy engineering studies before determining the incentive level.

In proposing this solution, VGIC recognizes that there are likely some aspects of this mechanism that may take additional time and stakeholder input to resolve. Some relevant questions are provided below, though VGIC does not propose specific answers to these at this time:

- What customer types are eligible for an ALM incentive (e.g. MUD, public charging, workplace charging, fleets)? Should there be any different treatment for customers participating in TE programs?
- Which specific avoided cost components would be linked to each incentive component (upfront, versus ongoing bill credit)?
- How should the bill credit be calculated (e.g. How are incremental distribution upgrade costs identified? Should it be linked to circuit-specific loading levels? Could a similar method be used to determine the “circuit adder” included in SDG&E’s Power Your Drive VGI Rate?)
- How should the bill credit be adjusted over time if additional distribution upgrades are ultimately needed to accommodate EV charging?
- What percent of the total savings should be shared with the ALM-provider/host customer and what percent should be retained by all ratepayers? Should an additional share of

savings be offered to customers pursuing ALM-enabled EVSE in underserved communities?

#### **IV. Timeline for Implementation**

VGIC believes that the basic implementation of AB 841 can and should proceed in a timely manner to establish the relevant tariffs required under the law even before any ALM incentive mechanisms is fully resolved. However, we also believe that it would be beneficial to revise those tariffs at a later date to incorporate the ALM incentive component being described here. Some additional time is likely needed to develop the details of this component (including the questions posed above), which need not be rushed. As such VGIC recommends that any Commission Order on AB 841 include an additional process (e.g. workshop and comment period) for further developing the ALM incentive that can be incorporated into the tariffs and approved in a subsequent Order.

#### **V. Implications for alleviating primary distribution costs via ALM through aggregation.**

While the approach being proposed here is targeted at the secondary distribution system upgrades associated with a single location, the general concept could be adapted in the future to using ALM to alleviate primary distribution system costs either at one location or across multiple locations. This could be performed by an aggregator such as an original equipment manufacturer (“OEM”) or EV Service Provider (“EVSP”) using load management in the same distribution grid footprint. Since AB 841 requires the Commission to consider implementation of new tariffs, VGIC recommends that such tariffs also consider including an aggregator option that would

perform this function. This may require additional stakeholder input beyond the immediate question of how to implement AB 841, but is worth considering as a longer-term objective that could leverage the Commission’s experience in this process.

## **VI. Response to Specific Questions in the ACR**

VGIC provides the following comments on specific sections of the January 15th ACR.

### *Common Treatment for Excess PEV (Plug-in Electric Vehicle) Charging*

VGIC agrees with the ACR’s interpretation of AB 841 and proposal that “the policy known as the Common Treatment for Excess PEV Charging continue indefinitely until revised by the Commission.”<sup>3</sup>

### *Cost Tracking*

The ACR sought comments on information the Commission would need to collect “with respect to cost data and other metrics, to determine the effectiveness of electrical corporation spending and promote transparency.”<sup>4</sup> VGIC supports collection of additional information on costs as proposed in the ACR. Additionally, if VGIC’s proposed shared savings mechanism is adopted, we recommend collecting data on the following: 1) total number of new EVSE installations that choose to include ALM and total number that do not include ALM, 2) total amount of distribution system upgrade cost avoided through ALM, 3) total portion of these

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<sup>3</sup> ACR at 3

<sup>4</sup> ACR at 6

savings (from 2) that were returned to ratepayers, and 4) total portion of these savings that were provided as an incentive to site hosts through the shared savings approach. 5) change in average utilization rates at EV installations.

### *Customer Contributions & Determining Reasonableness*

The ACR requests comment on the interpretation of AB 841 with respect to non-residential customer contributions. VGIC believes the plain language of AB 841 broadly exempts customers from being required to make contributions to utility-side costs, even beyond the current practice of granting allowances under Rules 15 and 16. As such, we do not think that overly prescriptive limitations such as those listed on pages 9-10 of the ACR are consistent with AB 841.

That said, we support having some general guidelines of what practices are acceptable to help ensure utility-side costs are kept reasonable, as long as these practices are largely consistent with current practice and do not place burdensome new restrictions on site host customers seeking to install EVSE. It is our understanding that such guidelines may be under development from the IOUs and could be a good starting point for further evaluation.

VGIC is particularly concerned about item 7 on page 10 of the ACR which proposes to cap the ratepayer cost contribution based on site host characteristics such as “the number and capacity of EV charging ports.” Given the wide range of potential customer charging needs and preferences, VGIC strongly recommends against any approach that would limit flexibility for new EV customers, such as a capacity-based cap on ratepayer cost contributions. While we recognize that techniques like ALM may make these limitations more manageable for some customers, we do not think a “one size fits all” approach is appropriate since it could deter new

EVSE deployments at a time when there is still a significant gap in necessary deployments to meet the state’s EV goals. As an alternative, the Commission could implement a shared savings mechanism like that proposed by VGIC above as a means to minimize ratepayer costs through a “carrot” rather than a “stick” approach. As the market matures, a more stringent approach could be considered. However, if any ratepayer cost contribution is capped (which, to reiterate, does not appear permissible under AB 841) on the basis of EV charging capacity, it should be done so at a sufficiently high enough level of capacity that it does not become a *de facto* ALM requirement.

With regards to item 4 (“requiring cost-sharing for the customer’s new meter”), VGIC urges the Commission to consider approaches (including the use of ALM), that could encourage the use of an existing utility service to accommodate new EVSE, thereby eliminating the need for a new separate service connection and meter. VGIC believes its proposed shared savings mechanism could go a long way towards accomplishing this goal.

#### *New Tariff, New Electric Rule, or Amended Electric Rules*

VGIC is agnostic as to whether the changes necessary for AB 841 should be accomplished through a new tariff, new rule, or amendment to an existing rule. However, we believe that whatever approach is adopted should be consistent across IOUs. Additionally, VGIC believes that any of these approaches provides an opportunity to develop a systematic process to implement the proposed shared savings mechanism described in these comments.

#### *Load Management Solutions*

The ACR seeks comment on “the question of whether the Commission should require the electrical corporations to offer load management solutions to customers taking advantage of the new rule or tariff required by AB 841.” VGIC believes that customers should not be *required* to implement load management solutions, since doing so could deter some customers from pursuing electrification. However, we support utilities being required to offer load management solutions to customers installing EVSE as a matter of course through TE programs or related infrastructure tariffs. The aim of ALM solutions in this context is to raise utilization rates of existing connections with no noticeable impact on customer experience. However, customers must have the choice of rejecting ALM entirely if they wish full nameplate service, or choosing an ALM solution if they see that their needs can reasonably be met while using such a system and could benefit from doing so. Furthermore, the shared savings mechanism for ALM that VGIC has proposed can and should be incorporated as a component of the AB 841-compliant tariffs as a means of fulfilling such a requirement, since it inherently encourages customers to pursue ALM as a load management solution. VGIC believes this would also be consistent with the requirements of D.20-12-029, which the ACR references: “any future tariff or rule filed by a large electrical corporation for service line and/or distribution line upgrades to support transportation electrification shall provide an option for customer side ALM [Automated Load Management] where beneficial to ratepayers while meeting TE charging needs.”<sup>5</sup>

With regards to implementation of other VGI services, the CPUC should look to implement ALM alongside a broader change in TE program design; namely, the need to consider utilization of existing utility service connections as a matter of course. This shift would ensure that the Commission meets the directives of Public Utilities Code Sec. 740.16 (a)(2) that “any

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<sup>5</sup> D.20-12-029, Section 6.1.

investments in transportation electrification do not foreclose the electric vehicle grid integration potential of these investments.” More specifically, many VGI use cases – such as optimization with solar PV or “vehicle-to-load” backup resiliency use cases – are contingent on having EV load intermingled with other host-site resources and loads. Put another way, allowing TE programs to be designed to require a separate service drop eliminates the potential for many high value VGI use cases to be achieved. Although use of ALM may reduce overall demand and thus could reduce the baseline available for measuring to grid-facing services like resource adequacy or distribution deferral, interconnecting under existing service will enable rather than inhibits a greater number of VGI use cases. Moreover, a reduced demand will achieve the same effect on the grid as achieving it through a demand response service.

Otherwise, the shared savings approach proposed by VGIC is voluntary and thus should eliminate any unintended consequences for customers that wish to pursue other VGI services. In contrast, requiring customers to implement load management as a condition of the AB 841 rule or tariff could have unintended consequences that would limit the VGI potential of the EVSE. For example, if customers were required to downsize their charging capacity through ALM, then that would also reduce their ability to maximize participation in future demand response or V2G activities since export capability would also be downsized. This could be especially detrimental in light of the rapidly growing potential for VGI to serve as a meaningful capacity resource to the grid during extreme weather events. If load management solutions are kept optional as VGIC has proposed, then customers can appropriately weigh the costs and benefits of pursuing an ALM incentive versus a higher discharge capacity for future VGI opportunities.

Respectfully submitted,

A handwritten signature in black ink, reading "Edward Burgess". The signature is fluid and cursive, with the first name "Edward" and last name "Burgess" clearly distinguishable.

Edward Burgess

Senior Policy Director

**VEHICLE-GRID INTEGRATION COUNCIL**

February 5, 2021